

A. M. ISMAIL

SYSTEM ARCHITECT | INDUSTRIAL AUTOMATION | FULL-STACK DEVELOPER

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PROFESSIONAL SUMMARY

Engineering student skilled in full-stack software development, hardware automation, and rapid prototyping with 3D printing. Backed with domain knowledge about manufacturing and supply chain and adept at building backend APIs, CLI tools, and self-hosted systems, with a strong drive to transform ideas into reliable, end-to-end hardware-software solutions. Eager to continuously learn, tackle challenging technical problems, and apply these skills to drive company growth.

KEY PROJECTS

Sri Energy Industrial Automation

[\[Project Details\]](#) | June 2025 – March 2026

Stack: ESP32, MQTT, FastAPI, React, PocketBase, NGINX, Docker, XAMPP

- Built an IoT control system for an overhead crane in a valve factory using ESP32 microcontrollers over MQTT, enabling workers to trigger instant emergency stops safely.
- Created a web dashboard in React with a FastAPI backend to give operators real-time control and machine monitoring options.

CTSKII ML Cloud Infrastructure

[\[Project Details\]](#) | Sept 2025 – March 2026

Stack: Docker Swarm, GlusterFS, JupyterHub, FastAPI, XAMPP, Let's Encrypt

- Set up a private GPU computing server by connecting 10 lab PCs with Docker Swarm so students could run and train machine learning models 24/7 for free.
- Built a web-based slot booking platform using FastAPI to let students easily schedule server time and launch GPU containers automatically.

SQLC – Quran Competition Management System

[\[Project Details\]](#) | June 2026

Stack: React, TypeScript, PocketBase, NGINX, Docker Compose

- Developed a full web application to manage event participant registrations, venue assignments, digital scoring marksheets, and real-time tie-resolution.
- Deployed the PocketBase database using Docker Compose to handle fast score updates smoothly during live event scoring.

EggShell – Visual Data Pipeline Builder

[\[Project Details\]](#) | May 2026

Stack: React, SQLite Web Worker, React Flow, OPFS, Web Workers

- Built a browser-based tool that lets users visually connect, filter, and clean spreadsheet data using an interactive React Flow canvas.
- Ran SQLite inside an in-browser Web Worker so users can query and merge large CSV/Excel files locally without uploading data to any external server.

CWM (Command Watch Manager)

[\[Project Details\]](#) | Nov 2025 – Dec 2025

Stack: Go (Golang), CLI Tooling, Cross-Compilation

- Built a fast, zero-dependency CLI tool compiled in Go to help developers save, tag, search, and execute long shell commands from a local history bank.
- Packaged the tool into a standalone single executable binary (.exe) with built-in utilities to jump between project directories and switch active GitHub profiles.

Virtual Lab for Metrology

[\[Project Live\]](#) | Sept 2025 – Apr 2026

Stack: React, HTML5, JavaScript, GitHub Pages

- Built web-based lab simulators for students to practice mechanical measuring tools (like Profile Projectors) interactively in the browser.
- Organized 12 experiment modules cleanly inside GitHub Pages repositories for free, direct access by students.

INTERNSHIP EXPERIENCE

SRI Energy Private Limited

June 2026

Operations & Systems Intern

Observed industrial valve assembly workflows, shop-floor logistics, and quality control steps.

CLUBS & EXTRACURRICULAR ACTIVITIES

Badminton Team Captain

Organized team practice, match lineups, and represented college in zonal tournaments.

2026

MITONAU Motorsports

Assisted go-kart team with basic frame assembly and event track logistics.

Dec 2024 – 2025

EDUCATION

Madras Institute of Technology (MIT), Anna University

Aug 2023 – 2027

Undergraduate Student

CGPA: 7.74 / 10

Higher Secondary Certificate (HSC)

Score: 545 / 600 (90.83%)

TECHNICAL SKILLS

Cad 3D modelling: SolidWorks, Catia, NX cad, Creo-Parametric, Abaqus.

Programming Languages: Go (Golang), Python, JavaScript, TypeScript, Dart, C++

Web & Mobile Development: React, FastAPI, Flutter, HTML/CSS, React Flow, Webhooks, REST APIs

Databases & Backend Tools: PocketBase, SQLite, Firebase, Supabase, Appwrite.

DevOps & Systems Tools: Docker, Docker Swarm, NGINX, MQTT, Linux/CLI Tooling, Git, GitHub Pages, XAMPP, Github Actions.